

WASP-50b

R-1635

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_140202 · created 2026-08-02T06:17:56+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456690.694 +/- 0.087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.049
Transit depth (Rp/Rs)^2	2.95 +/- 1.7 [%]
Orbital Inclination (inc)	88.02 +/- 2.95
Airmass coefficient 1 (a1)	0.842 +/- 0.058
Airmass coefficient 2 (a2)	0.117 +/- 0.047
Scatter in the residuals of the lightcurve fit is	6.87 %
Best Comparison Star	#2 - [105, 456]
Optimal Method	PSF photometry
Transit Duration (day)	0.029 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.172 ± 0.049 vs 0.1404 (deviation 0.64σ)
Duration fit vs published	0.029 d vs 0.0754 d (deviation 1.5σ)
Mid-time O–C	118.9 min
Depth SNR	3.5
Residual scatter	6.87 %

Light curve

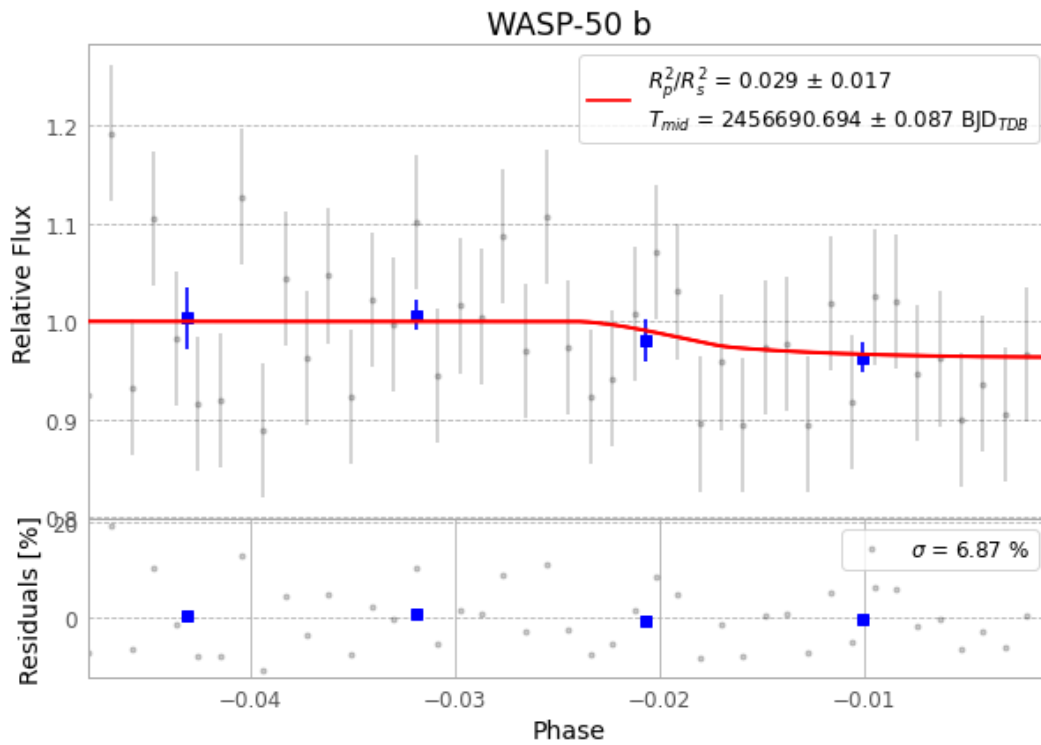


Figure 1 — FinalLightCurve_WASP-50 b_2014-02-01.png (SHA-256 8d90f7cb0f7a2513...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [210, 272], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d35a1897b3f640d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

45 FITS frames (30 MB), WASP-50140202022412.FITS → WASP-50140202043612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-140202.log (4765c768c69f...), FinalLightCurve_WASP-50 b_2014-02-01.png (8d90f7cb0f7a...), FinalLightCurve_WASP-50 b_2014-02-01.csv (36414af167d7...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 06:17Z.